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Feature Article

Computation of the energy levels of large-amplitude low-frequency vibrations. Comparison of the prediagonalized harmonic basis and the prediagonalized distributed Gaussian basis

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Abstract

The formalism involved in the solution of the Schrödinger equation for two-dimensional vibrational potential energy surfaces for large-amplitude low-frequency motions is reviewed. The performance of two different bases, the prediagonalized harmonic basis (PHB) and the prediagonalized distributed Gaussian basis (PDGB), is investigated. The calculated energy levels obtained with the two basis sets are in excellent agreement with one another. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

The out-of-plane ring-vibrations of cyclic molecules are typically of low-frequency and large-amplitude and are not well described by harmonic potential functions. The transitions involving these motions often give rise to very rich spectra, consisting not only of the fundamental bands, but also numerous hot bands, overtones and combination bands.

These low-frequency large-amplitude vibrations cannot adequately be treated by conventional normal coordinate methods. They are treated separately from the high frequency vibrations of the molecule. The degrees of freedom corresponding to the low-frequency motions are described using coordinates such as ring-twisting and ring-bending with the potential energy expressed as a polynomial in terms of these coordinates. Because the "real" vibrations often involve two or more of the chosen vibrational coordinates, one has to solve the vibrational problem on a multidimensional potential energy surface. The objective of the calculations is to determine a representation of the potential energy surface which optimally reproduces the observed energy levels of these vibrations.

Over the past three decades we have examined the vibrations for numerous cyclic and bicyclic molecules and determined their potential energy functions, which were typically one- or two-dimensional. Thus a wealth of information concerning the conformations and structures of these molecules has been gained [1–5]. For two-dimensional studies the Schrödinger equation for the vibrational motions was solved using a prediagonalized harmonic basis (PHB) [6]. In the 1980s Light and co-workers introduced the distributed Gaussian basis (DGB) [7–10]. We have now

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investigated the use of the prediagonalized distributed Gaussian basis (PDGB) for the calculations of relevance to our research. The purpose of the present paper is to review the formalism involved in these calculations and to compare the performance of the PHB and PDGB.

1.1. The vibrational Hamiltonian

The Hamiltonian of the internal motions of the molecule expressed in the center of mass system is given by Refs. [6,11]

$$H_{\mathrm{v}} = -\frac{1}{2}\hbar^2 \sum_{\mathrm{i=1}}^{\mathrm{3N-6}} \sum_{\mathrm{j=1}}^{\mathrm{3N-6}} \left(\frac{\partial}{\partial q_i} + \frac{1}{4}\frac{\partial \ln g}{\partial q_i} \right) \times g_{ij} \left(\frac{\partial}{\partial q_j} - \frac{1}{4}\frac{\partial \ln g}{\partial q_j} \right) + V(q_1, q_2, \ldots, q_{3N-6}) \tag{1.1}$$

where q_i and q_j are vibrational coordinates, g_{ij} are elements of the matrix **G** and g is the determinant of the matrix **G**.

The **G** matrix is of the form

$$\mathbf{G} = \begin{pmatrix} \mathbf{I} & \mathbf{X} \\ \mathbf{X}^{\mathrm{t}} & \mathbf{Y} \end{pmatrix}^{-1}, \tag{1.2}$$

where **I** is the 3×3 moment of inertia tensor, **X** the $3 \times (3N-6)$ matrix which represents the vibration–rotation interaction and **Y** the symmetric $(3N-6) \times (3N-6)$ matrix which describes the vibrations.

The matrix **G** written out for the case of two vibrations is

$$\mathbf{G} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} & X_{11} & X_{12} \\ -I_{yx} & I_{yy} & -I_{yz} & X_{21} & X_{22} \\ -I_{zx} & -I_{zy} & I_{zz} & X_{31} & X_{32} \\ X_{11} & X_{21} & X_{31} & Y_{11} & Y_{12} \\ X_{12} & X_{22} & X_{32} & Y_{21} & Y_{22} \end{pmatrix}^{-1} \tag{1.3}$$

where

$$I_{ii} = \sum_{\alpha=1}^{N} m_\alpha \left(\mathbf{r}_\alpha \cdot \mathbf{r}_\alpha - \mathrm{r}_{\alpha i}^2 \right), \qquad i = x, y, \text{ or } z \tag{1.4a}$$

$$I_{ik} = \sum_{\alpha=1}^{N} m_\alpha r_{\alpha i} r_{\alpha k}, \qquad i \neq k \tag{1.4b}$$

$$X_{ik} = \sum_{\alpha=1}^{N} m_\alpha \left[\mathbf{r}_\alpha \times \left(\frac{\partial \mathbf{r}_\alpha}{\partial q_k} \right) \right]_i \tag{1.4c}$$

$$Y_{ik} = \sum_{\alpha=1}^{N} m_\alpha \left(\frac{\partial \mathbf{r}_\alpha}{\partial q_i} \right) \cdot \left(\frac{\partial \mathbf{r}_\alpha}{\partial q_i} \right) \tag{1.4d}$$

The **G** matrix is of the form

$$\mathbf{G} = \begin{pmatrix} g_{11} & g_{12} & g_{13} & g_{14} & g_{15} \\ g_{21} & g_{22} & g_{23} & g_{24} & g_{25} \\ g_{31} & g_{32} & g_{33} & g_{34} & g_{35} \\ g_{41} & g_{42} & g_{43} & g_{44} & g_{45} \\ g_{51} & g_{52} & g_{53} & g_{54} & g_{55} \end{pmatrix}. \tag{1.5}$$

The description of the kinetic energy of the vibrations involves the elements g_{44}, g_{45}, and g_{55}. Because of the often rather large-amplitude of the motion, the dependence of the reduced masses on the vibrational coordinates has to be accounted for [12,13]. This is achieved by expressing g_{44}, g_{45}, and g_{55} as polynomials in the relevant coordinates:

$$g_{ij} = \sum_{k=0}^{m} \sum_{k=0}^{n} g_{ij}^{(k,l)} q_4^k q_5^l, \qquad i,j = 4,5; \; m + n \leq 6. \tag{1.6}$$

If

$$g_{ii} \approx g_{ii}^{(0,0)} \tag{1.7}$$

then

$$g_{ii} = \frac{1}{\mu_i}, \tag{1.8}$$

where μ is the reduced mass of vibration i ($i = 4, 5$). Here the numbering has been changed because the indices $i = 1, 2, 3$ are used to describe the rotations of the molecule.

The computation of the coefficients of the polynomials which describe the coordinate dependence of the elements of the matrix **G** is straightforward and has been described earlier [12,13]. The results depend, of course, on the molecular geometry and on the definitions and number of the vibrational coordinates used.

The kinetic energy part of the Hamiltonian can be divided into two parts, which in the case of two vibrations are

$$T_{\mathrm{v}} = -\frac{1}{2}\hbar^2\left(\frac{\partial}{\partial q_4}g_{44}\frac{\partial}{\partial q_4} + \frac{\partial}{\partial q_4}g_{45}\frac{\partial}{\partial q_5} + \frac{\partial}{\partial q_5}g_{54}\frac{\partial}{\partial q_4} + \frac{\partial}{\partial q_5}g_{55}\frac{\partial}{\partial q_5}\right) + V', \quad (1.9)$$

where

$$V' = \frac{1}{8}\hbar^2\sum_{i,j=4}^{5}\left(\frac{\partial}{\partial q_i}g_{ij}\frac{\partial \ln g}{\partial q_j} - \frac{\partial \ln g}{\partial q_i}g_{ij}\frac{\partial}{\partial q_j}\right) + \frac{1}{16}\hbar^2\sum_{i,j=4}^{5}\frac{\partial \ln g}{\partial q_i}g_{ij}\frac{\partial \ln g}{\partial q_j} \quad (1.10)$$

is called the pseudo potential. It is small compared to the potential V and usually omitted [14].

In the following we will denote the two coupled vibrational coordinates by x and y, the kinetic energy operator is then written as

$$2\hat{\mathbf{K}} = \hat{\mathbf{p}}_x g_{44}(x,y)\hat{\mathbf{p}}_x + \hat{\mathbf{p}}_y g_{55}(x,y)\hat{\mathbf{p}}_y + \hat{\mathbf{p}}_x g_{45}(x,y)\hat{\mathbf{p}}_y + \hat{\mathbf{p}}_y g_{45}(x,y)\hat{\mathbf{p}}_x \quad (1.11)$$

where

$$\hat{\mathbf{p}}_x = -i\hbar\frac{\partial}{\partial x}; \qquad \hat{\mathbf{p}}_y = -i\hbar\frac{\partial}{\partial y} \quad (1.12)$$

and

$$g_{ij}(x,y) = \sum_{m,n} g_{ij}^{(m,n)}x^m y^n;\ \ m+n \le 6; \qquad g_{44}^{(0,0)} = \frac{1}{\mu_x^0};\ \ g_{55}^{(0,0)} = \frac{1}{\mu_y^0} \quad (1.13)$$

The potential energy operator is written as

$$\hat{\mathbf{U}}(x,y) = \hat{\mathbf{U}}_1(x) + \hat{\mathbf{U}}_2(y) + \hat{\mathbf{U}}_{12}(x,y), \quad (1.13a)$$

where

$$\hat{\mathbf{U}}_1 = \sum_{i=1}^{6} C_{i0}\hat{\mathbf{x}}^i \qquad \text{and} \qquad \hat{\mathbf{U}}_2 = \sum_{j=1}^{6} C_{0j}\hat{\mathbf{y}}^j \quad (1.13b)$$

represent the potentials of one-dimensional vibrations, and

$$\hat{\mathbf{U}}_{12} = \sum_{i,j} C_{ij}\hat{\mathbf{x}}^i\hat{\mathbf{y}}^j; \qquad i+j \le 6 \quad (1.13c)$$

the coupling between the two vibrations. The one-dimensional eigenvectors, can be determined in the minimum of U, in which case $(\hat{\mathbf{U}}_1)_{\mathrm{eff}}(x) = \hat{\mathbf{U}}_1(x) + \hat{\mathbf{U}}_{12}(x, y_{\min})$ and $(\hat{\mathbf{U}}_2)_{\mathrm{eff}}(y) = \hat{\mathbf{U}}_2(y) + \hat{\mathbf{U}}_{12}(x_{\min}, y)$.

A typical potential function representation of the potentials we model is of the form

$$U(x,y) = A_1x^4 + B_1x^2 + A_2y^4 + B_2y^2 + Cx^2y^2 \quad (1.14)$$

If $B_1 < 0$ the function $U(x,0)$ is a double minimum potential with minima at $x = \pm\sqrt{B_1/2A_1}$. The height of the potential barrier between the minima is $\Delta U = B_1^2/4A_1$.

1.2. Optimization of the potential energy surface

The objective of the analysis is to determine the parameters of the potential energy function. This is accomplished by an iterative minimization of the differences between observed and calculated frequencies

$$|\Delta\boldsymbol{v}| = |\boldsymbol{v}_{\mathrm{obs}} - \boldsymbol{v}_{\mathrm{calc}}|, \quad (1.15)$$

where $|\Delta\boldsymbol{v}|$ is a column vector that contains all observed frequencies and the corresponding calculated frequencies. $\mathbf{F}$ is a column vector of the potential energy function parameters. The changes $\Delta\mathbf{F}$ to the parameters in each step are calculated as

$$\Delta\mathbf{F} = (\mathbf{J}^{\mathrm{t}}\mathbf{W}\mathbf{J})^{-1}\mathbf{J}^{\mathrm{t}}\mathbf{W}\Delta\boldsymbol{v} \quad (1.16)$$

$\mathbf{W}$ is a diagonal matrix ($n_{\mathrm{obs}} \times n_{\mathrm{obs}}$) containing the weights assigned to the observed frequencies on the diagonal and $\mathbf{J}$ is the Jacobian matrix ($n_{\mathrm{obs}} \times n_{\mathrm{par}}$) whose elements are

$$J_{ij} = \frac{\partial\Delta\boldsymbol{v}_i}{\partial F_j} = \langle\mathbf{f}_i|\frac{\partial U}{\partial F_j}|\mathbf{f}_i\rangle - \langle\mathbf{i}_i|\frac{\partial U}{\partial F_j}|\mathbf{i}_i\rangle. \quad (1.17)$$

Here $|i_i\rangle$ and $|f_i\rangle$ initial and final states corresponding to the observed transition with frequency ν_{i}. The computed intensities of the transitions are often helpful when correlating calculated energy levels with the spectroscopically determined ones. The intensities are proportional to

$$I(\upsilon_{f\leftarrow i})\alpha\left(\exp\left(-\frac{E_i hc}{kT}\right) - \exp\left(-\frac{E_f hc}{kT}\right)\right)\upsilon\langle\mathbf{f}|\mathrm{Z}|\mathbf{i}\rangle \tag{1.18}$$

where $Z = x, y, x^2, y^2,$ or xy.

2. Solving the Schrödinger equation

The Schrödinger equation of the vibrational motion

$$\hat{\mathbf{H}}|\psi_n\rangle = E_n|\psi_n\rangle, \tag{2.1}$$

where $\hat{\mathbf{H}} = \hat{\mathbf{K}} + \hat{\mathbf{U}}$, E_n is the nth vibrational energy, and $|\psi_n\rangle$ the corresponding wavefunction is solved by matrix diagonalization. The matrix elements are determined in a suitable finite basis. Two different choices of basis functions will be discussed, the PHB and the prediagonalized distributed Gaussian basis (PDGB).

2.1. The prediagonalized basis

When a two-dimensional Schrödinger equation is solved, it is most efficient to use a prediagonalized basis, where the two-dimensional basis functions are product functions of the eigenfunctions of two one-dimensional Hamiltonians. The eigenfunctions of the two-dimensional problem are then given by

$$|\Psi_n(x,y)\rangle = \sum_{i=0}^{i_{\max}}\sum_{j=0}^{j_{\max}} c_{ij}^{(n)}|\varphi_i^{(x)}\rangle|\varphi_i^{(y)}\rangle \tag{2.2a}$$

where the prediagonalized basis functions are defined by the equations

$$\hat{\mathbf{H}}_x|\varphi_i^x\rangle = E_i^{(x)}|\varphi_i^x\rangle; \qquad \hat{\mathbf{H}}_y|\varphi_j^y\rangle = E_j^{(y)}|\varphi_j^y\rangle \tag{2.2b}$$

where

$$\hat{\mathbf{H}}_x = \frac{1}{2}\hat{\mathbf{P}}_x^2 + \frac{1}{2}\sum_{m=1}^{6}\frac{g_{44}^{(m,0)}}{(\mu_x^0)^{(m-2)/2}}\,\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_x + \sum_{i=1}^{6}\frac{C_{i0}}{(\mu_x^0)^{i/2}}\hat{\mathbf{Q}}_x^i, \tag{2.2c}$$

and

$$\hat{\mathbf{H}}_y = \frac{1}{2}\hat{\mathbf{P}}_y^2 + \frac{1}{2}\sum_{n=1}^{6}\frac{g_{55}^{(0,n)}}{(\mu_y^0)^{(n-2)/2}}\hat{\mathbf{P}}_y\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y + \sum_{j=1}^{6}\frac{C_{0j}}{(\mu_y^0)^{j/2}}\hat{\mathbf{Q}}_y^j, \tag{2.2d}$$

The second term on the right hand side in these equation must sometimes left out from the one-dimensional analysis because of spurious unphysical wavefunctions and energies appearing in the solution of the Schrödinger equation. The reason is probably that the higher wavefunctions of the one-dimensional basis probe regions of coordinate space outside the region of validity of the reduced mass polynomial.

The vibrational Hamiltonian is formulated in mass-weighted vibrational coordinates

$$Q = \sqrt{\mu^0}q; \qquad P = \frac{p}{\sqrt{\mu^0}}; \qquad P = -i\hbar\frac{\partial}{\partial Q} \tag{2.3}$$

where μ^0 is the zeroth order term of the reduced mass of the vibrational coordinate.

The equations are solved by diagonalizing the matrices, calculated either in a harmonic or a DGB:

$$(H_x)_{km} = \langle u_m|\hat{H}_x|u_k\rangle; \qquad (H_y)_{ln} = \langle v_n|\hat{H}_y|v_l\rangle \tag{2.4}$$

This is achieved by the similarity transformation

$$\mathbf{A}^{-1}\mathbf{HA} = \mathrm{E}, \tag{2.5}$$

where $\mathbf{A}$ is a matrix whose columns are the eigenvectors and $\mathbf{E}$ is a diagonal matrix with the eigenvalues on the diagonal. We write the one-dimensional basis functions as

$$|\varphi_i^x\rangle = \sum_{k=0}^{k_{\max}} a_{ik}|u_k\rangle; \qquad |\varphi_j^y\rangle = \sum_{l=0}^{l_{\max}} b_{jl}|v_l\rangle. \tag{2.6}$$

The two-dimensional basis is then

$$|\varphi_i^x\rangle|\varphi_j^y\rangle = \sum_{k=0}^{k_{\max}}\sum_{l=0}^{l_{\max}} a_{ik}b_{jl}|u_k\rangle|v_l\rangle; \tag{2.7}$$

$i = 0,\ldots,i_{\max}\ \ j = 0,\ldots,j_{\max}$.

The full Hamiltonian is written as

$$\hat{\mathbf{H}} = \hat{\mathbf{H}}_x + \hat{\mathbf{H}}_y + \hat{\mathbf{H}}_{xy}, \tag{2.8}$$

where $\hat{\mathbf{H}}_{xy}$ contains all terms not included in the one-dimensional Hamiltonians. The kinetic energy part of the full Hamiltonian is

$$\hat{\mathbf{K}} = \frac{1}{2}\sum_{m,n=0}^{6} \frac{g_{44}^{(m,n)}}{(\mu_x^0)^{(m-2)/2}(\mu_y^0)^{(n/2)}} \hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_y^n + \frac{1}{2}\sum_{m,n=0}^{6} \frac{g_{55}^{(m,n)}}{(\mu_x^0)^{m/2}(\mu_y^0)^{(n-2)/2}} \hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_y\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y + \frac{1}{2}\sum_{m,n=0}^{6} \frac{g_{45}^{(m,n)}}{(\mu_x^0)^{(m-1)/2}(\mu_y^0)^{(n-1)/2}} \times\left[\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y + \hat{\mathbf{P}}_y\hat{\mathbf{Q}}_x^m\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_x\right]. \tag{2.9}$$

The commutator equality

$$[\hat{\mathbf{P}}_\alpha, \hat{\mathbf{Q}}_\alpha^m] = -\mathrm{i}\hbar m\hat{\mathbf{Q}}_\alpha^{m-1} \tag{2.10}$$

where α stands for x or y, can be used either to eliminate $\hat{\mathbf{Q}}_\alpha^m\hat{\mathbf{P}}_\alpha$ from the last bracket of Eq. (2.9), or preferably, to render it in the form

$$\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y + \hat{\mathbf{P}}_y\hat{\mathbf{Q}}_x^m\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_x = (1/2)\left[\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m + \hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_x\right] \times\left[\hat{\mathbf{P}}_y\hat{\mathbf{Q}}_y^n + \hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y\right] + (1/2)\hbar^2 mn\hat{\mathbf{Q}}_x^{m-1}\hat{\mathbf{Q}}_y^{n-1} \tag{2.11}$$

The sums in the brackets have antisymmetric matrix representations.

The matrix elements of the two-dimensional Hamiltonian are of the general form

$$\langle i'j'|\hat{\mathbf{O}}_x\hat{\mathbf{O}}_y|ij\rangle = \sum_k\sum_{k'}\sum_l\sum_{l'} a_{ik}a_{i'k'}b_{jl}b_{j'l'}\langle u_{k'}|\hat{\mathbf{O}}_x|u_k\rangle\langle v_{l'}|\hat{\mathbf{O}}_y|v_l\rangle. \tag{2.12}$$

The matrix can be constructed as the direct product of two matrices, each involving one coordinate:

$$(\mathbf{H})_{ij,i'j'} = (\mathbf{H}^{(x)})_{ii'} \otimes (\mathbf{H}^{(y)})_{jj'}, \tag{2.13a}$$

where

$$(\mathbf{H}^{(x)})_{ii'} = \langle\varphi_i^{(x)}|\hat{\mathbf{H}}_x|\varphi_{i'}^{(x)}\rangle; \qquad (\mathbf{H}^{(y)})_{jj'} = \langle\varphi_j^{(y)}|\hat{\mathbf{H}}_y|\varphi_{j'}^{(y)}\rangle \tag{2.13b}$$

The positions of the elements in the matrix are computed as

$$ij = (i-1)j_{\max} + j; \qquad i'j' = (i'-1)j_{\max} + j'; \quad ,i' = 1,\dots,i_{\max},\ j,j' = 1,\dots,j_{\max} \tag{2.13c}$$

where $i_{\max}$ and $j_{\max}$ are the number of one-dimensional basis functions $\{|\varphi_i^{(x)}\rangle\}$ and $\{|\varphi_i^{(y)}\rangle\}$ included in the product basis.

The Hamiltonian $\hat{\mathbf{H}}_{xy}$ contains the following terms:

1. The cross-terms in the potential function gives

$$\frac{C_{m,n}}{(\mu_x^0)^{m/2}(\mu_y^0)^{n/2}}\hat{\mathbf{Q}}_x^m\hat{\mathbf{Q}}_y^n; \quad 1 \le m \le 6;\ 1 \le n \le 6;\ m+n \le 6 \tag{2.14a}$$

2. If the one-dimensional eigenfunctions are computed in the minimum of the potential, then the terms added to $(U_1)_{\mathrm{eff}}$ and $(U_2)_{\mathrm{eff}}$ have to be subtracted:

$$-\frac{C_{m,n}y_{\min}^n}{(\mu_x^0)^{m/2>}}\hat{\mathbf{Q}}_x^m; \qquad -\frac{C_{m,n}x_{\min}^m}{(\mu_y^0)^{n/2}}\hat{\mathbf{Q}}_y^n; \quad 1 \le m \le 6,\ 1 \le n \le 6,\ m+n \le 6 \tag{2.14b}$$

3. The dependence of the reduced mass of one vibrational mode on the values of the other vibrational coordinate gives rise to the following terms:

$$\frac{1}{2}\frac{g_{44}^{(m,n)}}{(\mu_x^0)^{(m-2)/2}(\mu_y^0)^{n/2}}\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_y^n; \quad 0 \le m \le 6,\ 1 \le n \le 6,\ m+n \le 6 \tag{2.14c}$$

and

$$\frac{1}{2}\frac{g_{55}^{(m,n)}}{(\mu_x^0)^{m/2}(\mu_y^0)^{(n-2)/2}}\hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_y\hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y; \quad 1 \le m \le 6,\ 0 \le n \le 6,\ m+n \le 6 \tag{2.14d}$$

4. The cross term in the kinetic energy expansion gives the following terms:

5.

$$\frac{1}{4}\frac{g_{45}^{(m,n)}}{(\mu_x^0)^{(m-1)/2}(\mu_y^0)^{(n-1)/2}}\Big[(\hat{\mathbf{P}}_x\hat{\mathbf{Q}}_x^m + \hat{\mathbf{Q}}_x^m\hat{\mathbf{P}}_x)(\hat{\mathbf{P}}_y\hat{\mathbf{Q}}_y^n + \hat{\mathbf{Q}}_y^n\hat{\mathbf{P}}_y) + \hbar^2 mn\hat{\mathbf{Q}}_x^{m-1}\hat{\mathbf{Q}}_{\mathrm{y}}^{\mathrm{n}-1}\Big]$$

$$0 \le m \le 6, \qquad 0 \le n \le 6, \qquad m + n \le 6. \tag{2.14e}$$

2.1.1. The harmonic basis

The harmonic wavefunctions that form the basis of the one-dimensional computations are defined by the equations

$$\hat{\boldsymbol{H}}_0^{(x)}|u_k\rangle = E_k|u_k\rangle; \qquad \hat{\mathbf{H}}_0^{(x)} = \frac{1}{2}\hat{\mathbf{P}}_x^2 + \omega_x^2\hat{\mathbf{Q}}_x^2; \qquad E_k = \left(k + \frac{1}{2}\right)\hbar\omega_x \tag{2.15a}$$

and

$$\hat{\mathbf{H}}_0^{(y)}|v_l\rangle = E_l|v_l\rangle; \qquad \hat{\mathbf{H}}_0^{(y)} = \frac{1}{2}\hat{\mathbf{P}}_y^2 + \omega_y^2\hat{\mathbf{Q}}_y^2; \qquad E_l = \left(l + \frac{1}{2}\right)\hbar\omega_y \tag{2.15b}$$

The harmonic frequencies ω_x and ω_y can usually be chosen within rather large bounds. In practice one has to use finite bases. Usually $k_{\max} = 60$, $l_{\max} = 60$, $i_{\max} = 10$ and $j_{\max} = 10$ suffice.The matrix elements needed to build the Hamiltonian operator are of the following types:

$$\langle k|\hat{\mathbf{Q}}^n|l\rangle = \langle u_k|\mathbf{Q}^n|\mathbf{u}_l\rangle \text{ with } 1 \le n \le 6$$

$$\langle k|\hat{\mathbf{P}}\hat{\mathbf{Q}}^n\hat{\mathbf{P}}|l\rangle - \hbar^2\langle u_k|\frac{\partial}{\partial Q}Q^n\frac{\partial}{\partial Q}|u_l\rangle$$

with $0 \le n \le 6$

$$\langle k|\hat{\mathbf{P}}\hat{\mathbf{Q}}^n + \hat{\mathbf{Q}}^n\hat{\mathbf{P}}|l\rangle - \mathrm{i}\hbar\langle u_k|\frac{\partial}{\partial Q}Q^n + Q^n\frac{\partial}{\partial Q}|u_l\rangle$$

with $0 \le n \le 6$ (2.16)

The matrices $(\hat{\mathbf{Q}}^n)_{kl}$ and $(\hat{\mathbf{P}}\hat{\mathbf{Q}}^{\mathrm{n}}\hat{\mathbf{P}})_{kl}$ are symmetric with respect to permutation of the indices k and l, whereas the matrices $(\hat{\mathbf{P}}\hat{\mathbf{Q}}^n + \hat{\mathbf{Q}}^n\hat{\mathbf{P}})_{kl}$ are anti-symmetric.

The derivation of these matrix elements is straightforward using the raising and lowering operators $a^\dagger$ and a, where

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \qquad a|n\rangle = \sqrt{n}|n-1\rangle. \tag{2.17}$$

The position operator $\hat{\mathbf{Q}}$ and momentum operator $\hat{\mathbf{P}}$ can be expressed in terms of these:

$$\hat{\mathbf{Q}} = \sqrt{\frac{\hbar}{2\omega}}(a^\dagger + a) \qquad \hat{\mathrm{P}} = \mathrm{i}\sqrt{\frac{\hbar\omega}{w}}(a^\dagger - a), \tag{2.18}$$

where $i = \sqrt{-1}$.

Most of the matrix elements have been tabulated in a paper by Harthcock and Laane [6], but there are some errors in those tables, in particular for the PQ elements. Therefore we reproduce them here for easy reference. In the expressions below

$$\delta = \sqrt{\frac{\hbar}{2\omega}} = \sqrt{\frac{h}{8\pi^2 c\bar{\upsilon}}} \qquad \text{and } \sigma = \sqrt{\frac{\hbar\omega}{2}} = \sqrt{\frac{hc\bar{\upsilon}}{2}} \tag{2.19}$$

where $\bar{\upsilon}$ is the harmonic wavenumber chosen for the basis.

$$\langle n|Q|n+1\rangle = \delta(n+1)^{1/2} \tag{2.20a}$$

$$\langle n|Q|n\rangle = 0$$

$$\langle n|Q^2|n+2\rangle = \delta^2[(n+1)(n+2)]^{1/2}$$

$$\langle n|Q^2|n\rangle = \delta^2(2n+1)$$

$$\langle n|Q^3|n+3\rangle = \delta^3[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|Q^3|n+1\rangle = 3\delta^3(n+1)^{3/2}$$

$$\langle n|Q^4|n+4\rangle = \delta^4[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|Q^4|n+2\rangle = \delta^4(4n+6)[(n+1)(n+2)]^{1/2}$$

$$\langle n|Q^4|n\rangle = \delta^4(6n^2 + 6n + 3)$$

$$\langle n|Q^5|n+5\rangle = \delta^5[(n+1)(n+2)(n+3)(n+4)(n+5)]^{1/2}$$

$$\langle n|Q^5|n+3\rangle = 5\delta^5(n+2)[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|Q^5|n+1\rangle = 5\delta^5(2n^2+4n+3)(n+1)^{1/2}$$

$$\langle n|Q^6|n+6\rangle = \delta^6[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)]^{1/2}$$

$$\langle n|Q^6|n+3\rangle = 3\delta^5(2n+5)[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|Q^6|n+2\rangle = 15\delta^6(n^2+3n+3)[(n+1)(n+2)]^{1/2}$$

$$\langle n|Q^6|n\rangle = 5\delta^6(4n^3+6n^2+8n+3)$$

$$\langle n|P^2|n+2\rangle = -\sigma^2[(n+1)(n+2)]^{1/2} \tag{2.20b}$$

$$\langle n|P^2|n\rangle = \sigma^2(2n+1)$$

$$\langle n|PQP|n+3\rangle = -\sigma^2\delta[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQP|n+1\rangle = \sigma^2\delta(n+1)^{3/2}$$

$$\langle n|PQ^2P|n+4\rangle = -\sigma^2\delta^2[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|PQ^2P|n+2\rangle = 0$$

$$\langle n|PQ^2P|n\rangle = \sigma^2\delta^2(2n^2+2n+3)$$

$$\langle n|PQ^3P|n+5\rangle = -\sigma^2\delta^3[(n+1)(n+2)(n+3)(n+4)(n+5)]^{1/2}$$

$$\langle n|PQ^3P|n+3\rangle = -\sigma^2\delta^3(n+2)[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQ^3P|n+1\rangle = \sigma^2\delta^3(2n^2+4n+9)(n+1)^{1/2}$$

$$\langle n|PQ^4P|n+6\rangle = -\sigma^2\delta^4[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)]^{1/2}$$

$$\langle n|PQ^4P|n+4\rangle = -\sigma^2\delta^4(2n+5)[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|PQ^4P|n+2\rangle = \sigma^2\delta^4(n^2+3n+15)[(n+1)(n+2)]^{1/2}$$

$$\langle n|PQ^4P|n\rangle = \sigma^2\delta^4(4n^3+6n^2+32n+15)$$

$$\langle n|PQ^5P|n+7\rangle = -\sigma^2\delta^5[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)(\mathrm{n}+7)]^{1/2}$$

$$\langle n|PQ^5P|n+5\rangle = -\sigma^2\delta^5(3n+9)[(n+1)(n+2)(n+3)(n+4)(n+5)]^{1/2}$$

$$\langle n|PQ^5P|n+3\rangle = -\sigma^2\delta^5(n^2+4n-15)[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQ^5P|n+1\rangle = \sigma^2\delta^5(5n^3+15n^2+85n+75)[n+1]^{1/2}$$

$$\langle n|PQ^6P|n+8\rangle = -\sigma^2\delta^6[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)(n+7)(n+8)]^{1/2}$$

$$\langle n|PQ^6P|n+6\rangle = -\sigma^2\delta^6(4n+14)[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)]^{1/2}$$

$$\langle n|PQ^6P|n+4\rangle = -4\sigma^2\delta^6 n(n+5)[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|PQ^6P|n+2\rangle = 2\sigma^2\delta^5(2n^3+9n^2+79n+105)[(n+1)(n+2)]^{1/2}$$

$$\langle n|PQ^6P|n\rangle = 5\sigma^2\delta^5(2n^4+4n^3+46n^2+44n+21)$$

$$\langle n|P|n+1\rangle = -\mathrm{i}\sigma(n+1)^{1/2} \tag{2.20c}$$

$$\langle n|PQ+QP|n+2\rangle = -\mathrm{i}\sigma\delta[(n+1)(n+2)]^{1/2}$$

$$\langle n|PQ^2+Q^2P|n+3\rangle = -2i\sigma\delta^2[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQ^2+Q^2P|n+1\rangle = -2i\sigma\delta^2(n+1)(n+1)^{1/2}$$

$$\langle n|PQ^3+Q^3P|n+4\rangle = -2i\sigma\delta^3[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|PQ^3+Q^3P|n+2\rangle = -i\sigma\delta^3(4n+6)[(n+1)(n+2)]^{1/2}$$

$$\langle n|PQ^4+Q^4P|n+5\rangle = -2i\sigma\delta^4[(n+1)(n+2)(n+3)(n+4)(n+5)]^{1/2}$$

$$\langle n|PQ^4+Q^4P|n+3\rangle = -6i\sigma\delta^4(n+2)[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQ^4+Q^4P|n+1\rangle = -i\sigma\delta^4(4n^2+8n+6)(n+1)^{1/2}$$

$$\langle n|PQ^5+Q^5P|n+6\rangle = -2i\sigma\delta^5[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)]^{1/2}$$

$$\langle n|PQ^5+Q^5P|n+4\rangle = -i\sigma\delta^5(8n+20)[(n+1)(n+2)(n+3)(n+4)]^{1/2}$$

$$\langle n|PQ^5+Q^5P|n+2\rangle = -i\sigma\delta^5(10n^2+30n+30)[(n+1)(n+2)]^{1/2}$$

$$\langle n|PQ^6+Q^6P|n+7\rangle = -2i\sigma\delta^6[(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)(n+7)]^{1/2}$$

$$\langle n|PQ^6+Q^6P|n+4\rangle = -i\sigma\delta^6(10n+30)[(n+1)(n+2)(n+3)(n+4)(n+5)]^{1/2}$$

$$\langle n|PQ^6+Q^6P|n+3\rangle = -i\sigma\delta^6(18n^2+72n+90)[(n+1)(n+2)(n+3)]^{1/2}$$

$$\langle n|PQ^6+Q^6P|n+1\rangle = -i\sigma\delta^6(10n^3+30n^2+50n+30)[n+1]^{1/2}$$

2.2. The distributed Gaussian basis

The DGB provides an alternative to the (prediagonalized) harmonic basis. This basis has been thoroughly described by Hamilton and Light [7]. It allows flexibility in choosing the coverage of the potential surface by basis functions and is useful when a non-polynomic representation of the potential is used.

2.2.1. The one-dimensional problem

The basis functions are Gaussian functions

$$\Phi_i(x) = \left(\frac{2A_i}{\pi}\right)^{\frac{1}{4}} \exp\left[-A_i(x - x_i)^2\right] \tag{2.21}$$

distributed over the potential function. These basis functions are not linearly independent, however. They have to be orthogonalized prior to the diagonalization of the Hamiltonian matrix.

The centers x_i of these functions are chosen as described by Hamilton and Light. The idea is the following:

The nth bound state wave function has $n - 1$ nodes spaced approximately semiclassically, i.e. if x_i is the position of the ith node, the classical action between x_i and x_{i+1} is

$$\Delta J_n = \int_{x_i}^{x_{i+1}} p(x)\,\mathrm{d}x = \frac{h}{2} \tag{2.22}$$

where p is the classical momentum of a particle with energy E_n at x and h is Planck's constant. The momentum $p(x)$ is calculated as

$$p(x) = \sqrt{2\mu(x)[E_n - U(x)]}, \tag{2.23}$$

where $\mu(x)$ is the coordinate dependent reduced mass and $U(x)$ is the potential function. Since the wave functions are to be approximated by linear combinations of Gaussians, we need at least one basis function centered between each node of the highest wave function we want to calculate correctly. We chose the positions of these Gaussians as the x_i which satisfy the conditions

$$\int_{x_{\min}}^{x_1} p(x)\,\mathrm{d}x = h/8; \qquad \int_{x_i}^{x_{i+1}} p(x)\,\mathrm{d}x = h/2 \tag{2.24}$$

For a particle moving with the bound state energy E_{n} (where we define the lowest state energy as E_1) the number n of this energy state is given by

$$R_L = \frac{2}{h}\int_{x_{\min}}^{x_{\max}} p(x)\,\mathrm{d}x + \frac{1}{2} = n. \tag{2.25}$$

Here $x_{\min}$ is the left turning point of the classical bound particle moving in the potential with energy E_n, and $x_{\max}$ is the right turning point.

In order to calculate the E_n state with sufficient accuracy more basis functions have to be used. These are distributed in the semiclassical manner described above simply by rescaling the parameter h as

$$h \to h\left(R_L - \frac{1}{2}\right)/\left(M - \frac{1}{2}\right), \tag{2.26}$$

where M is the number of Gaussians in the basis and R_L is computed for some energy higher than the highest state E_n that is to be computed correctly. The exact positions of the Gaussians are not important. The interval ($x_{\min}$, $x_{\max}$) is divided into a large number of intervals of equal length, and the position of the mth Gaussian is then defined by the value of x at which the integral of pdx first becomes greater than $h/8 + (m - 1)h/2$.

The widths of the basis functions are chosen such that a suitable compromise between low kinetic energy (small A_i) and a reasonable linear independence (large A_i) is achieved. The coefficients A_i are chosen in such a way as to keep the overlap of the functions moderate, i.e. larger where the distribution is denser:

$$A_i = 4c^2/(x_{i+1} - x_{i-1})^2;$$
$$A_1 = c^2/(x_2 - x_1)^2; \tag{2.27}$$
$$A_N = c^2/(x_N - x_{N-1})^2$$

The parameter c can be chosen rather freely. Hamilton and Light give the range $0.5 - 1.1$. For $c = 0.5$ the Gaussian has dropped to $\approx e^{-0.25} = 0.78$ of its maximum value at the center of its adjacent Gaussians. Larger values for c make the Gaussians narrower and cause ripples in the wavefunctions. Smaller values again increase the overlap between the basis functions, and cause numerical difficulties in the orthogonalization of the basis. The calculations described in the next section indicate, that the optimum value for this parameter is close to $c = 0.5$.

The matrix elements of the Hamiltonian operator can be computed analytically in this basis if the potential function is represented by a polynomial in the vibrational coordinates. Also the matrix elements of exponential and Gaussian potential terms can be determined analytically. However, the eigenvalue problem has to be expressed in terms of the symmetrically orthogonalized basis before diagonalisation of the Hamiltonian. This is accomplished by multiplying the Hamiltonian matrix **H** symmetrically by the inverse square root of the overlap matrix **S**:

$$\tilde{\mathbf{H}} = \mathbf{S}^{-\frac{1}{2}}\mathbf{H}\mathbf{S}^{-\frac{1}{2}} \tag{2.28}$$

The elements S_{ij} of the matrix **S** are defined by the integrals

$$S_{ij} = \int_{\infty -}^{\infty} \Phi_i \Phi_j \,\mathrm{d}x \tag{2.29}$$

.

2.2.2. Two-dimensional potentials

The DGP is easily generalized to more than one dimension. In the two-dimensional problem the basis functions are of the form

$$\begin{aligned}\Psi_i(x,y) &= \sqrt{\frac{2A_i}{\pi}}\exp\left[-A_i(r-r_i)^2\right] \\ &= \sqrt{\frac{2A_i}{\pi}}\exp\left[-A_i(x-x_i)^2\right]\exp\left[-A_i(y-y_i)^2\right] \\ &= \Phi_i(x)\Phi_i(y)\end{aligned} \tag{2.30}$$

It is easily seen that the two-dimensional overlap integral is the product of the two one-dimensional overlap integrals

$$S_{ij} = S_{ij}^{(x)} S_{ij}^{(y)} \tag{2.31}$$

All matrix element in the Gaussian basis are obtained as the product of two integrals, one with respect to x, the other with respect to y. The calculation of the matrix elements in the two-dimensional case is thus equivalent to forming the products of two one-dimensional calculations. The difficulty compared to the one-dimensional case is, that the size of the basis will be much larger and therefore it is important to optimize the distribution of the centers of the Gaussian basis functions.

Light and Hamilton [7] distributed the two-dimensional basis functions evenly over the classically allowed region of the potential energy surface for some chosen total energy, using when possible the symmetry properties of the Hamiltonian to reduce the size of the matrices involved. We have chosen here to use a simple-minded extension to two dimensions of the one-dimensional distribution, by making the nearest neighbor distances between the centers approximately inversely proportional to $\sqrt{E-V}$.

The selection of the Gaussian centers is done in the following way, which should be applicable to an arbitrary two-dimensional potential surface for an energy at which the motion is classically restricted in all directions:

1. A dense 200×200 grid of points is placed on a rectangle centered at the origin and with its corners at ($\pm x_{\max}$, $\pm y_{\max}$), where $x_{\max}$ and $y_{\max}$ are the maximum coordinate values classically accessible by the vibrations at energy E.
2. Then each of these grid-points is considered in turn. A Gaussian basis function ψ_i is placed on the grid-point, if the point is in the classically allowed region at energy E and the square of the distance to the nearest neighbor Gaussian already placed on the grid is larger than a parameter D, where

$$D = \frac{\int_A (E - V(x,y))\,\mathrm{d}A}{N(E - V(x_i,y_i))} \tag{2.32}$$

The coefficient A_i of Eq. (2.44a) and (2.44b) is then set to $A_i = c^2/R_i^2$, where R_i is the distance to the nearest neighbor already on the grid. Here A is the classically allowed region of the potential energy surface at energy E, and N is the total number of basis functions that are to be placed on the surface.

The actual number of basis functions placed on the surface by the computer code using the criterion in step 2 is about 10% smaller than the parameter N, which is provided by the operator. This scheme probably does not produce the optimal basis, but it is straightforward and works in the general case.

A more efficient approach to the two-dimensional problem is, however, to use prediagonalized basis functions, as outlined above in Section 2.1. The one-dimensional problems are solved using the basis described in Section 2.2.1. The orthogonalized matrices are constructed for all operators in the two-dimensional Hamiltonian displayed in Eqs. (2.14a), (2.14b), (2.14c), (2.14d) and (2.14e) and the two-dimensional Hamiltonian is constructed using Eqs. (2.13a), (2.13b) and (2.13c).

The computation of the eigenvalues and the eigenfunctions occurs in three steps:

1. The Hamiltonian **H** and the overlap matrix **S** are computed in the chosen non-orthogonal basis.
2. The eigenvalue problem is expressed in the symmetrically orthogonalised basis:

$$\mathbf{S}^{(-1/2)}\mathbf{H}\mathbf{S}^{(-1/2)} - \lambda\mathbf{I} = \tilde{\mathbf{H}} - \lambda\mathbf{I} = 0 \tag{2.33}$$

3. The eigenvalue problem is solved by diagonalization of $\tilde{\mathbf{H}}$.

Step 1 involves the evaluation of the matrix elements displayed in Eqs. (2.41a), (2.41b) and (2.41c) and is relatively fast. Step 2 is quite time consuming, as the matrices involved are large.

If the potential is being optimized, the changes to the parameters of the potential energy surface are computed in a fourth step. This requires that the matrices $(\hat{x}^m)_{ij}$ and $(\hat{y}^n)_{ij}$ are orthogonalized separately and stored in memory, so that the Jacobian matrix elements (Eq. (1.17)) can be evaluated. These matrices are also needed for the computation of the transition elements given in Eq. (1.18). Fortunately this time-consuming step has to be done only once, since only the potential parameters are changed after each iteration, the basis being kept unchanged. Of course, the classically allowed region of the potential surface also changes somewhat, but if it is determined for high enough energy this does not matter too much. It is usually the first few energy states that are involved in the optimization, and they are not very much affected by this.

If we write the eigenfunction corresponding to state i obtained in step 3 as

$$\Theta_i = \sum_{j=1}^{N} c_{ij}\Psi_j \tag{2.34}$$

then the elements of the Jacobian matrix (Eq. (1.17)) are given by expressions involving terms such as

$$\int \Theta_i x^m y^n \Theta_i \, \mathrm{d}x\,\mathrm{d}y = \sum_{j=1}^{N}\sum_{k=1}^{N} c_{ij}c_{ik}(\hat{x}^m)_{jk}(\hat{y}^n)_{jk}$$

$$= \left(\hat{Z}^{(m,n)}\right)_{ii}, \tag{2.35}$$

where $(\hat{x}^m)_{jk}$ and $(\hat{y}^m)_{jk}$ are the matrix elements shown in Eqs. (2.41a), and

$$\hat{\mathbf{Z}}^{(m,n)} = \mathbf{C}\hat{\mathbf{z}}^{(m,n)}\mathbf{C}^{-1} \tag{2.36}$$

Here $(\hat{z}^{(m,n)})_{jk} = (\hat{x}^m)_{jk}(\hat{y}^n)_{jk}$. The transition elements given in Eq. (1.18) are proportional to the off-diagonal elements of these matrices.

In the optimization of the potential only steps 3 and 4 need to be performed in each cycle of the process.

2.2.3. The matrix elements of the gaussian basis

The matrix elements of the Hamiltonian operator can be computed analytically in the Gaussian basis if the potential function is represented by a polynomial in the vibrational coordinates. Also the matrix elements of exponential and Gaussian potential terms can be determined analytically.

The overlap matrix **S** is defined by the integrals

$$S_{ij} = \int_{-\infty}^{\infty} \Phi_i\Phi_j \, \mathrm{d}x = \sqrt{\pi}\frac{A_{ij}}{\sqrt{A_i + A_j}} \exp(-C_{ij}) \tag{2.37}$$

where the coefficients are defined as

$$A_{ij} = \left(\frac{4A_iA_j}{\pi^2}\right)^{1/4}; \qquad B_{ij} = \frac{A_iA_j}{A_i + A_j};$$
$$\text{and } C_{ij} = \frac{A_iA_j}{A_i + A_j}(x_i - x_j)^2. \tag{2.38}$$

We will also use the following quantities:

$$x_{ij} = \frac{A_ix_i + A_jx_j}{A_i + A_j}, \qquad E_{ij} = \frac{A_jx_i + A_ix_j}{A_i + A_j}$$
$$\text{and } D_{ij} = \frac{A_iA_j}{A_i + A_j}(x_i - x_j) \tag{2.39}$$

to express the integrals

$$(\hat{x}^n)_{ij} = \int \Phi_i x^n \Phi_j \,\mathrm{d}x; \qquad (\hat{p}\hat{x}^n\hat{p})_{ij} = -\hbar^2 \int \Phi_i \frac{\partial}{\partial x} x^n \frac{\partial}{\partial x} \Phi_j \,\mathrm{d}x; \qquad (\hat{p}\hat{x}^n)_{ij} = -\mathrm{i}h \int \Phi_i \frac{\partial}{\partial x} x^n \Phi_j \,\mathrm{d}x;$$

$$(\hat{x}^n\hat{p})_{ij} = -\mathrm{i}\hbar^2 \int \Phi_i x^n \frac{\partial}{\partial x} \Phi_j \,\mathrm{d}x \tag{2.40}$$

which form the Hamiltonian matrix. These integrals are obtained analytically:

$$(\hat{x})_{ij} = x_{ij}S_{ij} \qquad (\hat{x}^2)_{ij} = \left[x_{ij}^2 + \frac{1}{2(A_i + A_j)}\right]S_{ij} \qquad (\hat{x}^3)_{ij} = \left[x_{ij}^3 + \frac{3x_{ij}}{2(A_i + A_j)}\right]S_{ij} \tag{2.41a}$$

$$(\hat{x}^4)_{ij} = \left[x_{ij}^4 + \frac{3x_{ij}^2}{A_i + A_j} + \frac{3}{4(A_i + A_j)}\right]S_{ij} \qquad (\hat{x}^5)_{ij} = \left[x_{ij}^5 + \frac{5x_{ij}^3}{A_i + A_j} + \frac{15x_{ij}}{4(A_i + A_j)^2}\right]S_{ij}$$

$$(\hat{x}^6)_{ij} = \left[x_{ij}^6 + \frac{15x_{ij}^4}{2(A_i + A_j)} + \frac{45x_{ij}^2}{4(A_i + A_j)^2} + \frac{15}{8(A_i + A_j)^3}\right]S_{ij}$$

$$(\hat{p}^2)_{ij} = -2\hbar^2\Big[B_{ij}(1 - 2C_{ij})\Big]S_{ij} \qquad (\hat{p}\hat{x}\hat{p})_{ij} = -2\hbar^2 B_{ij}\Big[E_{ij} - 2x_{ij}(1 - C_{ij})\Big]S_{ij} \tag{2.41b}$$

$$(\hat{p}\hat{x}^2\hat{p})_{ij} = -\hbar^2 B_{ij}\left[\frac{(2C_{ij} - 3)}{A_i + A_j} + 2x_{ij}\Big[2E_{ij} + x_{ij}(2C_{ij} - 3)\Big]\right]S_{ij}$$

$$(\hat{p}\hat{x}^3\hat{p})_{ij} = -\hbar^2 B_{ij}\left[\frac{3E_{ij} + 6x_{ij}(C_{ij} - 2)}{A_i + A_j} + 2x_{ij}^2\Big[3E_{ij} + x_{ij}(2C_{ij} - 4)\Big]\right]S_{ij}$$

$$(\hat{p}\hat{x}^4\hat{p})_{ij} = -\hbar^2 B_{ij}\left[\frac{3(2C_{ij} - 5)}{2(A_i + A_j)^2} + \frac{6x_{ij}\Big[2E_{ij} + x_{ij}(2C_{ij} - 5)\Big]}{A_i + A_j} + 2x_{ij}^3\Big[4E_{ij} + x_{ij}(2C_{ij} - 5)\Big]\right]S_{ij}$$

$$(\hat{p}\hat{x}^5\hat{p})_{ij} = -\hbar^2 B_{ij}\left[\frac{15E_{ij} + 30x_{ij}(C_{ij} - 3)}{2(A_i + A_j)^2} + \frac{+10x_{ij}^2\Big[3E_{ij} + 2x_{ij}(C_{ij} - 3)\Big]}{A_i + A_j} + 2x_{ij}^4\Big[5E_{ij} + 2x_{ij}(C_{ij} - 3)\Big]\right]S_{ij}$$

$$(\hat{p}\hat{x}^6\hat{p})_{ij} = -\hbar^2 B_{ij}\left[\frac{15(2C_{ij} - 7)}{4(A_i + A_j)^3} + \frac{45x_{ij}\Big[2E_{ij} + x_{ij}(2C_{ij} - 7)\Big]}{2(A_i + A_j)^2} + \frac{15x_{ij}^3\Big[4E_{ij} + x_{ij}(2C_{ij} - 7)\Big]}{A_i + A_j}\right.$$

$$\left. +2x_{ij}^5\Big[6E_{ij} + x_{ij}(2C_{ij} - 7)\Big]\right]S_{ij}$$

The cross-terms in the kinetic energy operator require the following integrals:

$$(\hat{p})_{ij} = -i\hbar(-2D_{ij})S_{ij} \qquad (\hat{p}\hat{x} + \hat{x}\hat{p})_{ij} = -i\hbar\left[\frac{A_i - A_j}{A_i + A_j} - 4x_{ij}D_{ij}\right]S_{ij} \tag{2.41c}$$

$$(\hat{p}\hat{x}^2 + \hat{x}^2\hat{p})_{ij} = -i\hbar\left[\frac{2(A_i - A_j)x_{ij} - 2D_{ij}}{A_i + A_j} - 4x_{ij}^2D_{ij}\right]S_{ij}$$

$$(\hat{p}\hat{x}^3 + \hat{x}^3\hat{p})_{ij} = -i\hbar\left[\frac{3(A_i - A_j)}{2(A_i + A_j)^2} + \frac{3(A_i - A_j)x_{ij}^2 - 6x_{ij}D_{ij}}{A_i + A_j} - 4x_{ij}^3D_{ij}\right]S_{ij}$$

$$(\hat{p}\hat{x}^4 + \hat{x}^4\hat{p})_{ij} = -i\hbar\left[\frac{12xA_i - A_j)x_{ij} - 6D_{ij}}{2(A_i + A_j)^2} + \frac{4(A_i - A_j)x_{ij}^3 - 12x_{ij}^2D_{ij}}{A_i + A_j} - 4x_{ij}^4D_{ij}\right]S_{ij}$$

$$(\hat{p}\hat{x}^5 + \hat{x}^5\hat{p})_{ij} = -i\hbar\left[\frac{15(A_i - A_j)}{4(A_i + A_j)^3} + \frac{30(A_i - A_j)x_{ij}^2 - 30x_{ij}D_{ij}}{2(A_i + A_j)^2} + \frac{5(A_i - A_j)x_{ij}^4 - 20x_{ij}^3D_{ij}}{A_i + A_j} - 4x_{ij}^5D_{ij}\right]S_{ij}$$

$$(\hat{p}\hat{x}^6 + \hat{x}^6\hat{p})_{ij} = -i\hbar\left[\frac{90(A_i - A_j)x_{ij} - 30D_{ij}}{4(A_i + A_j)^3} + \frac{60(A_i - A_j)x_{ij}^3 - 90x_{ij}^2D_{ij}}{2(A_i + A_j)^2} + \frac{6(A_i - A_j)x_{ij}^5 - 30x_{ij}^4D_{ij}}{A_i + A_j}\right.$$
$$\left. -4x_{ij}^6D_{ij}\right]S_{ij}$$

The matrix elements of an exponential term in the potential are easily obtained as

$$(\mathrm{e}^{-\omega x})_{ij} = S_{ij}\,\mathrm{e}^{-F_{ij}};$$
$$F_{ij} = \omega x_{ij} - \frac{\omega^2}{4(A_i + A_j)} \tag{2.42}$$

A Gaussian potential of the form

$$U(x) = -\exp\left[-\omega(x \pm x_0)^2\right] \tag{2.43}$$

has matrix elements of the form

$$(U)_{ij} = -S_{ij}\exp\left[-G_{ij}\right]\sqrt{\frac{A_i + A_j}{A_i + A_j + \omega}} \tag{2.44a}$$

where

$$G_{ij} = \omega x_0^2 + (A_i + A_j)x_{ij}^2 - \frac{\left[x_{ij}(A_i + A_j) \mp \omega x_0\right]^2}{\omega + A_i + A_j}. \tag{2.44b}$$

3. Comparison of the harmonic and the distributed Gaussian bases

In this section we investigate how the calculated energies depend on basis size and on the choice of harmonic frequency of the harmonic basis or the parameter c, which determines the extent of the overlap of adjacent Gaussians in the Gaussian basis. Two Hamiltonians are used, a simple harmonic potential with coordinate independent reduced mass, and the Hamiltonian of the ring-puckering and ring-flapping motions of phthalan [15].

3.1. Harmonic potential

The potential energy function (energy in cm^{-1} and position in Å) is

$$V = 59320.31461x^2 \tag{3.1}$$

(not all digits shown) with $\mu = 625$ amu, which gives a wavenumber of 80 cm^{-1} for the classical vibrational motion.

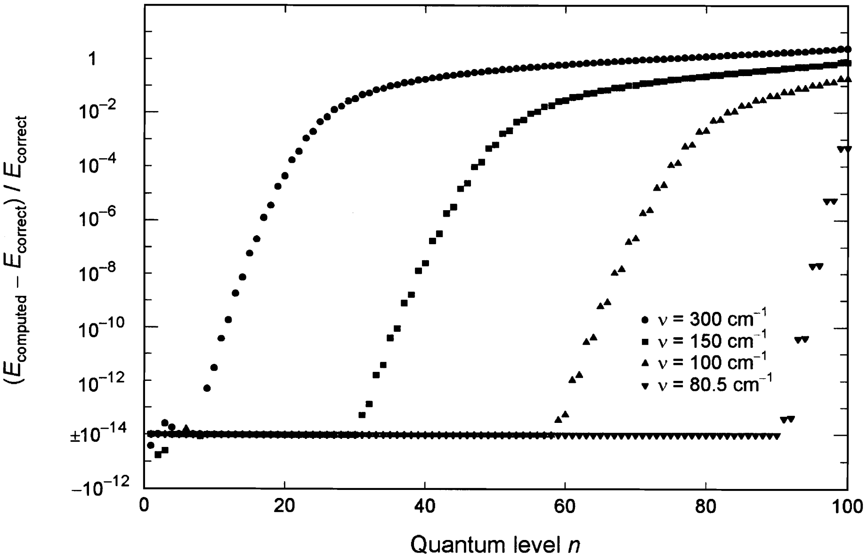

Fig. 1. Relative errors of the computed energy levels of a harmonic Hamiltonian with a classical oscillation of wavenumber 80 cm^{-1} computed with harmonic bases of different harmonic wavenumbers. The wavenumbers of the bases are indicated. Absolute values smaller than 10^{-14} are placed on the $\pm 10^{-14}$ line. Negative errors are shown below this line. Basis set size is $N = 100$.

3.1.1. Harmonic basis

Fig. 1 shows the effect of choosing a different wavenumber for the harmonic basis for a basis size of 100 basis functions. The relative errors of the computed energies are displayed as a function of energy level number, the numbering starting with $n = 1$. The numerical precision of the computation is of the order of 1 in 10^{15}. The deviations are plotted on a

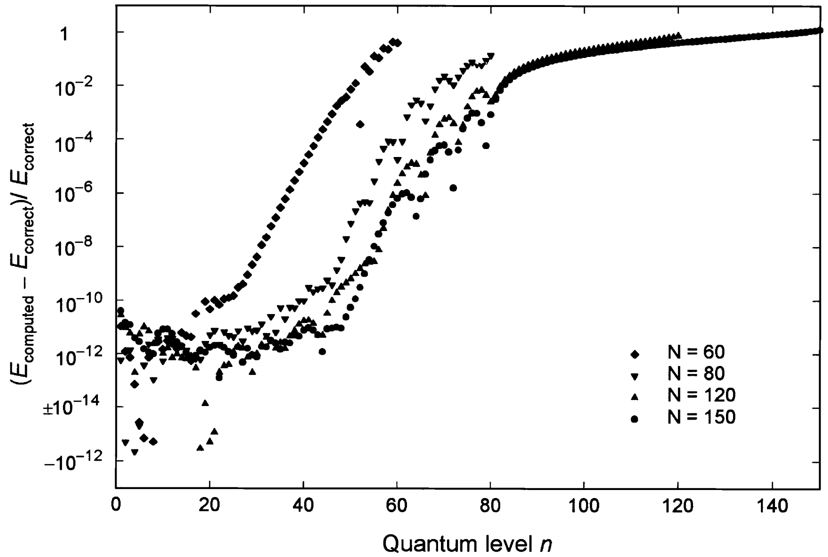

Fig. 2. Dependence on basis set size of the energy levels of a harmonic Hamiltonian ($\bar{\upsilon} = 80$ cm^{-1}) computed with a distributed Gaussian basis. The energy for which the Gaussians were positioned was 5000 cm^{-1}, corresponding to energy level $n = 63$ (the numbering starting from $n = 1$). $c = 0.50$. The sizes of the bases is indicated.

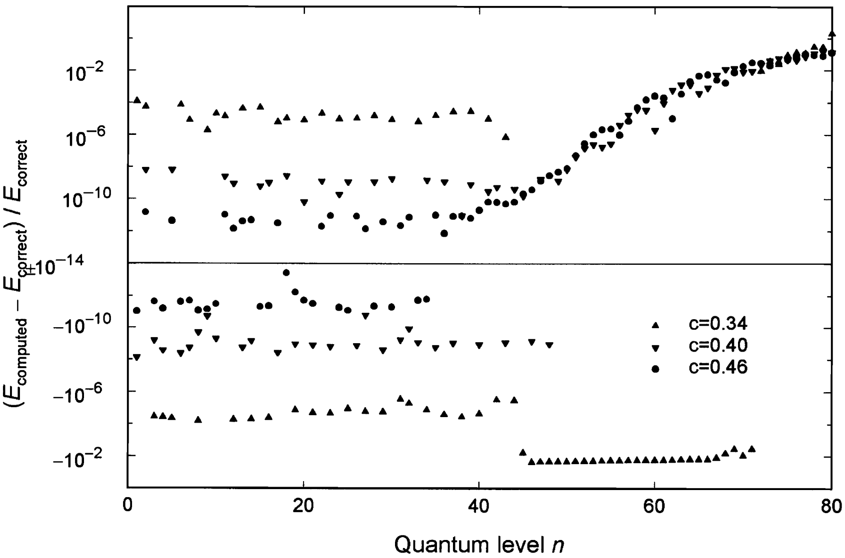

Fig. 3. Dependence of the computed energy levels of a harmonic Hamiltonian ($\bar{\upsilon} = 80\ \mathrm{cm}^{-1}$) on the parameter c of the distributed Gaussian basis. Basis set size is $N = 80$ and the energy for which the Gaussian functions were positioned is $E = 5000\ \mathrm{cm}^{-1}$ corresponding to energy level $n = 63$. The quantum level numbering starts from $n = 1$. The c-values are indicated.

logarithmic scale, with negative values below the $\pm10^{-14}$ line and all deviations with absolute value smaller than 10^{-14} set equal to that value.

The obvious choice of wavenumber for the harmonic basis is of course $80\ \mathrm{cm}^{-1}$, which would give all computed energies correctly. The further away from the optimum value the chosen harmonic frequency of the basis is, the fewer energy levels are correctly computed. Increasing the basis set size increases the number of correctly computed energy levels. We also find, that all energy levels converge to the same value with increasing basis set size, independently of the choice of harmonic frequency, with a precision better than 1 part in 10^{-14} for most levels. Only for the lowest few levels is the error of the computed energy slightly larger. Choosing too small a harmonic frequency for the basis causes a similar deviation of the computed energies as does choosing too large a frequency.

3.1.2. Distributed Gaussian basis

Three parameters determine the computed energy levels with the Gaussian basis, the basis size *N*, the energy *E* for which the positions of the basis functions have been determined, and the parameter *c*, which determines the overlap of the non-orthogonal Gaussian functions.

Fig. 2 shows the effect of the basis set size. The parameter $c = 0.50$ (Eq. 2.27) and $E = 5000\ \mathrm{cm}^{-1}$, corresponding to energy level $n = 63$ for all cases. It is seen, that increasing the basis size from $N = 80$ to $N = 150$ has a rather small effect on the number of converged energy levels, which is about 40. The errors in the computed values of the converged levels are both positive and negative and of the order of 1 part in 10^{12}. These errors arise in the orthogonalization of the basis. They increase with increasing overlap of the Gaussians, as is shown in Fig. 3. The orthogonalization failed for $c < 0.34$. Fig. 4 shows the effect of increasing the parameter *c*. These results clearly indicate, that there is an optimum value for the overlapping of the non-orthogonal Gaussians, which corresponds to $c \approx 0.5$. However, for practical purposes the accuracy of the converged energy levels is sufficient for $0.4 \leq c \leq 0.8$. More than half the computed energy levels are obtained with good accuracy.

Hamilton and Light [7] investigated the convergence of the computed energies with increasing basis set size, and found that the accuracy of even

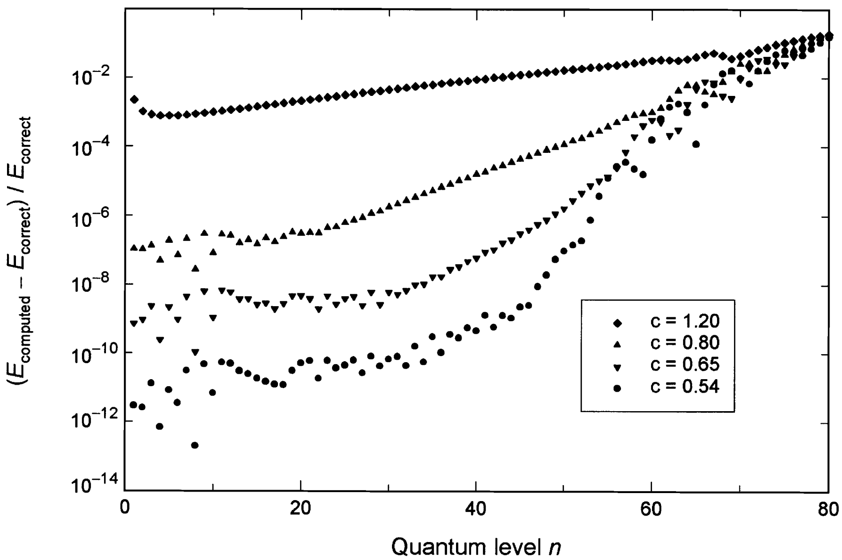

Fig. 4. Dependence of the computed energy levels of a harmonic Hamiltonian ($\bar{\upsilon} = 80\ cm^{-1}$) on the parameter c of the distributed Gaussian basis. Basis set size is $N = 80$ and the energy for which the Gaussian functions were positioned is $E = 5000\ cm^{-1}$corresponding to energy level $n = 63$. The numbering of the energy levels start from $n = 1$. The c-values are indicated.

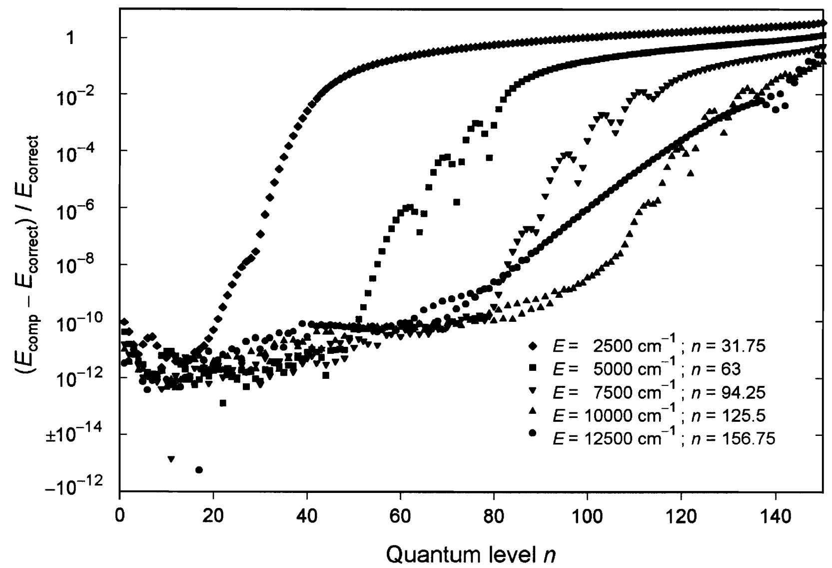

Fig. 5. Energy levels computed for a harmonic Hamiltonian ($\bar{\upsilon} = 80\ cm^{-1}$) using a Gaussian basis of size $N = 150$ and with $c = 0.50$. The effect of choosing the energy E for which the Gaussian functions are positioned is shown. The energies and corresponding level number are indicated in the figure. A non-integer value of n shows that the energy chosen does not correspond exactly to an eigenstate of the Hamiltonian.

the lowest energy states increased with increasing size. They, however, chose to keep the width of the Gaussian functions approximately constant by making c inversely proportional to the number of basis functions. What they observed, is really the effect on the computed energies of decreasing the parameter c.

The effect of varying the energy E for which the positions of the Gaussians are determined is shown by Fig. 5. The errors of the computed energies increase rapidly as a function of energy above the energy E. For this harmonic potential 120 energy levels are obtained with quite good accuracy with a basis of 150 Gaussians with $E = 10\,000\ \mathrm{cm}^{-1}$ corresponding to energy level $n = 125$. In these calculations one Gaussian basis function was placed outside the classically allowed region at each turning point of the classical motion, at a separation from the nearest neighbor equal to the distance between the two basis functions closest to the turning point in the classically allowed region.

3.2. The anharmonic potential function

The potential and kinetic terms of the Hamiltonian of phthalan [15] for its out-of-plane motions are shown in (Eqs. 3.2 and 3.3). They were determined from its far-infrared spectrum. The potential function is

$$\begin{aligned} V = {} & 34.8913038 - 8746.219407\, x^2 + 623364.2824\, x^4 \\ & + 6959.052054\, y^2 - 10441.90969 y^4 \\ & + 20733.59815\, y^6 + 4104.707058\, xy \\ & + 39488.68828 x^2 y^2 \end{aligned} \tag{3.2}$$

This potential has minima $V(x,y) = 0$ at $x = \pm 0.08639$ Å; $y = \mp 0.02448$ Å with a barrier of height $34.89\ \mathrm{cm}^{-1}$ at the origin between the minima. The potential energy parameters used here are the result of a different optimization from the one published in Ref. [15], but the function is essentially the same. The kinetic energy part of the Hamiltonian is

$$\begin{aligned} g_{44} = {} & 0.01051249 - 0.05612798 x^2 - 0.1224187 x^4 \\ & - 0.4656293 x^6 + 0.00438605 y^2 \\ & - 0.03086507 y^4 + 0.04979458 y^6 \\ & + 0.04133395 x^2 y^2 + 0.02102806 x^3 y \\ & + 0.005302694 x y^3 \\ g_{55} = {} & 0.11475654 - 0.09824962 y^2 + 0.05803999 y^4 \\ & - 0.05755477 y^6 + 0.1216271 x^2 - 1.655906 x^4 \\ & + 9.169819 x^6 \\ & - 0.05386573 x^2 y^2 + 0.1569516 x^3 y + 0.04409159 x y^3 \\ g_{45} = {} & -0.01669950 + 0.03006188 x^2 \\ & + 0.002971108 y^2 - 0.04505776 x^2 y^2 \\ & + 0.5853787 x^4 - 0.1666159 x^3 y \\ & - 0.04240851 x y^3 + 0.02645411 y^4 \\ & - 1.316742 x^6 - 0.0436674 y^6. \end{aligned} \tag{3.3}$$

In the expressions above x represents the puckering coordinate and y the flapping coordinate. The Hamiltonian is expressed in cm^{-1} units and the vibrational coordinates in angstroms. The one-dimensional Hamiltonians are obtained by setting $y = 0$ for the puckering function and $x = 0$ for the flapping function in the Eq. (3.2) and using only the zeroth order term of the kinetic energy functions g_{44} and g_{55} in Eq. (3.3), respectively. The reduced mass of the puckering motion is 95.125 amu and of the flapping mode 8.714 amu. The energy levels of the one-dimensional Hamiltonians and of the full two-dimensional Hamiltonian are displayed in Table 1. These were used [15] in the determination of the potential function Eq. (3.2). Table 1 also compares the two-dimensional levels to those expected for a molecule with no kinetic or potential energy interactions and where the coordinate dependence of the reduced masses is neglected.

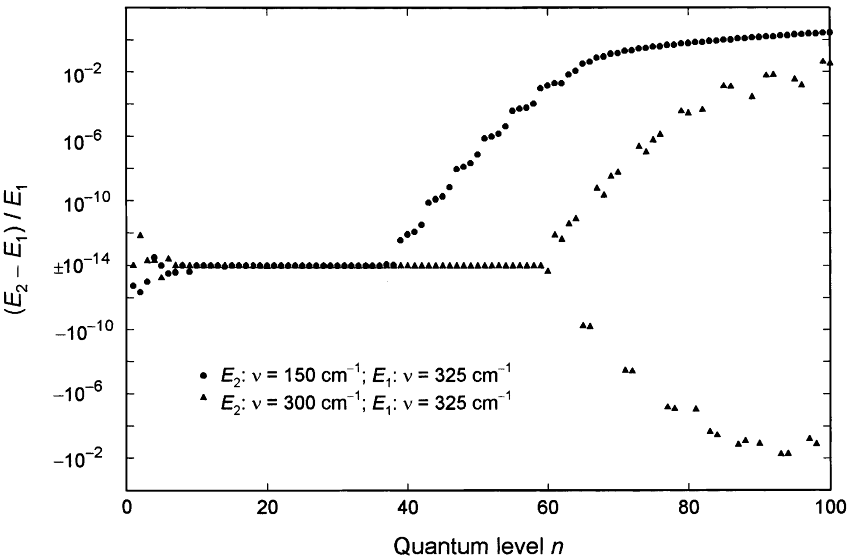

Fig. 6. Two comparisons of two one-dimensional computations (the puckering levels) for phthalan using bases with different harmonic wavenumbers. The harmonic wavenumbers are indicated. The basis set size is $N = 100$.

3.2.1. The harmonic basis

We begin by investigating the convergence behavior of the harmonic basis with respect to basis set size and the harmonic frequency chosen. The one-dimensional potential functions are not harmonic, and the task is to find the optimal harmonic frequency of the basis, which gives the lowest eigenvalues. We do this with two basis set sizes, $N = 60$ and $N = 100$. We start with some low wavenumber for the basis, calculate the eigenvalues, then increase the wavenumber and recalculate the eigenvalues. If the eigenvalues decrease, the change in wavenumber is taken as an improvement. We continue the process until the results no longer improve.

The results are illustrated by Fig. 6, which shows the relative differences of the eigenvalues of the puckering motion for two sets of harmonic wavenumbers, 325 cm^{-1} compared to 300 cm^{-1} and 325 cm^{-1} compared to 150 cm^{-1}. When the harmonic wavenumber of the basis is increased above 325 cm^{-1} the eigenvalues start to become larger and hence their errors become larger. The general behavior of the eigenvalues is the same as for the harmonic potential, in that they converge to a value, which is independent of the harmonic wavenumber and size of the harmonic basis. The number of eigenvalues that have converged, for a given basis set size, is largest for the optimal harmonic wavenumber of the basis. In the case shown 60 eigenvalues have converged with a basis size $N = 100$. Moreover, the optimum harmonic wavenumber of the basis seems to correspond to the energy level separation in the energy region of the highest converged levels. With the smaller basis set size $N = 60$, the optimum harmonic wavenumber of the basis is 225 cm^{-1} and the number of converged energy levels is 25. Similar results are obtained for the flapping mode potential, the number of converged energy levels being 50 (20) and the optimum harmonic wavenumber being 500 cm^{-1} (350 cm^{-1}) for the basis set size $N = 100$ ($N = 60$).

The result of the two-dimensional computation is independent of the sizes and harmonic wavenumbers of the bases used in the one-dimensional computations as long as the one-dimensional bases are large enough, so that the eigenvalues corresponding to the functions included in the two-dimensional basis are converged. The one-dimensional computations are relatively fast, so fairly large bases sets should be used to ensure the convergence of the basis functions used in the two-dimensional computation.

Figs. 7 and 8 illustrate the effect of changing the size of the two-dimensional basis. Our computer code

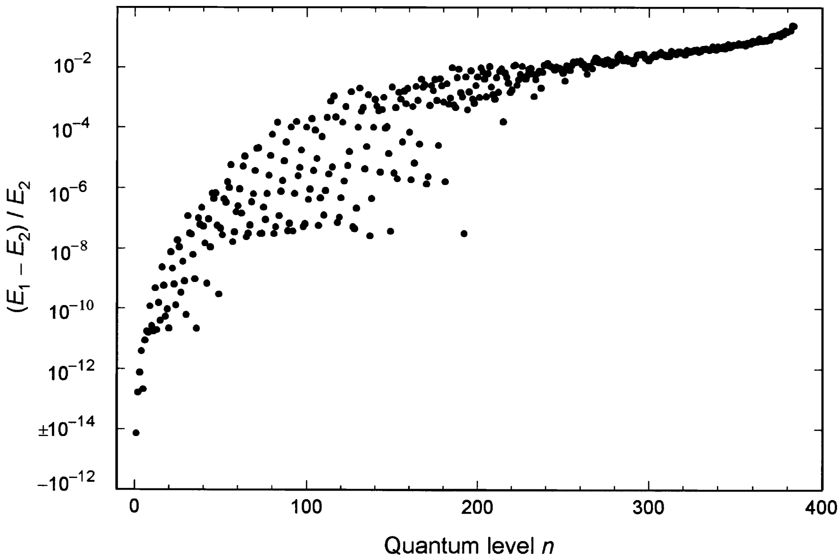

Fig. 7. Comparison of two two-dimensional computations of the energy levels of phthalan using prediagonalized harmonic bases of sizes 24 × 16 for E_2 and 25 × 16 for E_1. The one-dimensional bases were of sizes $N_1 = N_2 = 60$ with harmonic wavenumbers $\bar{\upsilon}_1 = 225\ \text{cm}^{-1}$ and $\bar{\upsilon}_2 = 375\ \text{cm}^{-1}$.

presently allows only 400 basis functions, which has been considered more than sufficient. Usually the computations have been carried out with a basis size of only $100(10 \times 10)$. Fig. 7 compares the results obtained with the basis sizes 25×16 and 24×16 and Fig. 8 the results for basis sizes 25×16 and 25×15. The different patterns of the deviations reflect the varying contributions from the puckering and flapping

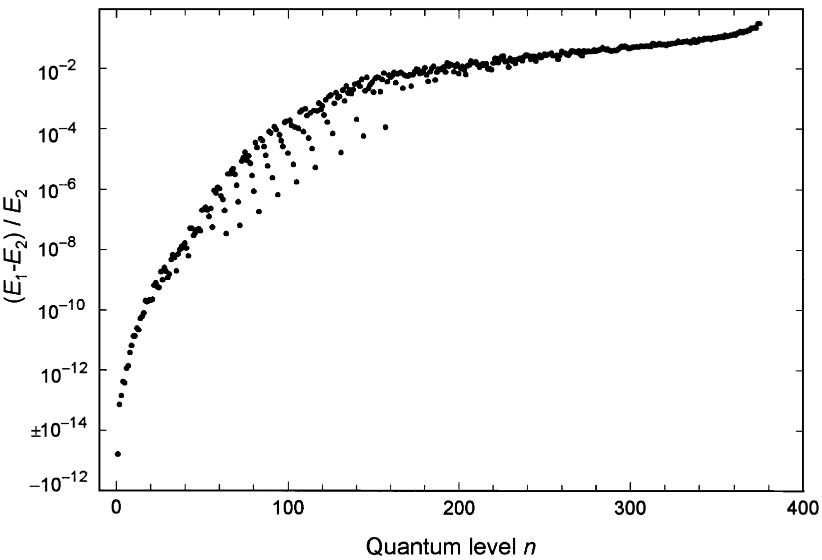

Fig. 8. Comparison of two two-dimensional computations of the energy levels of phthalan using prediagonalized harmonic bases of sizes 25 × 15 for E_2 and 25 × 16 for E_1. The one-dimensional bases were of sizes $N_1 = N_2 = 60$ with harmonic wavenumbers $\bar{\upsilon}_1 = 225\ \text{cm}^{-1}$ and $\bar{\upsilon}_2 = 375\ \text{cm}^{-1}$.

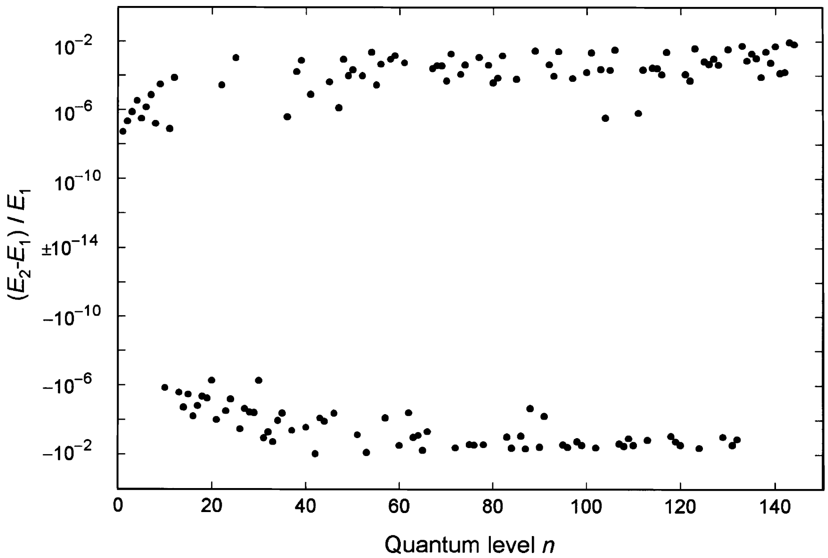

Fig. 9. Comparison of two two-dimensional computations of the energy levels of phthalan with prediagonalized harmonic bases of size 12 × 12. The parameters of the one-dimensional bases used in the computations were $N_1 = 60$, $\bar{v}_1 = 750$ cm^{-1}, $N_2 = 60$, $\bar{v}_2 = 375$ cm^{-1}, for E_2, and $N_1 = 60$, $\bar{v}_1 = 225$ cm^{-1}, $N_2 = 60$, $\bar{v}_2 = 375$ cm^{-1} for E_1.

modes. It is evident from the figures, and from calculations not shown, that the eigenvalues, except perhaps the first few, are not converged. However for all practical purposes the energy levels are obtained with sufficient accuracy. At most 50 energy levels would be experimentally determined and involved in construction of an empirical potential surface. Usually the number of energy levels experimentally determined is much smaller. More basis functions from the puckering motion were included in the two-dimensional basis, because the experimentally determined energy levels involve higher quanta of this vibration.

In Fig. 9 the results of two computations using bases of 12 × 12 are compared. Here the one-dimensional bases had the sizes $N = 60$. One computation was done using optimal harmonic wavenumbers for the bases, 225 and 375 cm^{-1}, respectively, whereas the other was done using the wavenumbers 750 and 375 cm^{-1}. The differences are both positive and negative reflecting the fact that neither computation has converged. However, even though the relative differences between the results of these two computations are relatively large, the computed energies of the spectroscopically determined levels differ even in this case at most by only a few hundredths of a wavenumber. These differences are inconsequential for the determination of the empirical potential function.

3.2.2. The prediagonalized Gaussian basis

In Figs. 10–12 the results obtained with a Gaussian basis are compared with the results obtained with the harmonic basis using the optimal wavenumbers 325 and 500 cm^{-1} for the puckering and flapping modes, respectively. The basis sizes in the one-dimensional computations was $N = 100$ for both bases and the two-dimensional bases both of size 25 × 16. The energies for which the Gaussians were positioned were 20 000 and 30 000 cm^{-1} for the two modes, corresponding to energy levels $n = 82$ and $n = 79$. The c-parameter of the Gaussian basis was chosen as $c = 0.50$. It is observed that the number of one-dimensional energy levels that have converged is about 50 for both modes, i.e. the harmonic and the Gaussian basis have performed equally well. The numerical precision of the convergence is slightly worse for the Gaussian basis, however, being of the order of 1 in 10^{12}. As shown by Fig. 12, there is complete agreement between the Gaussian and harmonic bases for the two-dimensional computations when the same number of one-dimensional basis functions is used.

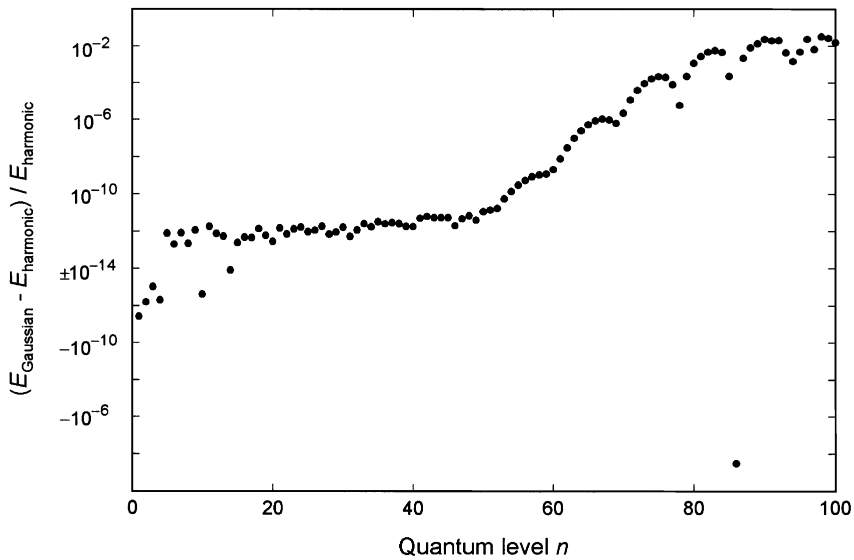

Fig. 10. Comparison of the energy levels of the puckering mode of phthalan calculated with a harmonic basis, $N = 100$, $\bar{\upsilon} = 325\ \mathrm{cm}^{-1}$, and a Gaussian basis, $N = 100$, $E = 20\,000\ \mathrm{cm}^{-1}$ ($n = 82$), $c = 0.50$.

The effect of changing the number of basis functions is much larger, as shown by Figs. 7 and 8.

3.2.3. The two-dimensional distributed Gaussian basis

Hamilton and Light [7] solved the Schrödinger equation for several two-dimensional potentials. Fig. 13 shows a comparison of the results obtained with the two-dimensional DGP, the first of size $N = 716$ with the Gaussian functions distributed as described in Section 2.2.2 above using $E = 8000\ \mathrm{cm}^{-1}$ corresponding approximately to energy level $n = 450$, the second of size $N = 708$ obtained using $E = 2500\ \mathrm{cm}^{-1}$ corresponding approximately to energy level $n = 138$, with the result obtained with the PDGB described above. The parameter $c = 0.50$ for all calculations. The effect of the energy E is quite large, the results obtained with $E = 2500\ \mathrm{cm}^{-1}$ being two orders of magnitude closer to the ones obtained with the PDGB. It is seen, that although the agreement with the result obtained with the PDGB is good for the first 50 energy states, the calculated energies are much further from convergence than the ones obtained with the much smaller PDGB. The precision seems also to be somewhat lower, as indicated by the difference obtained in the second computation for the first energy state (not shown in Fig. 13), which was -10^{-9}. The computations using the two-dimensional DGB are also much more time consuming.

4. Conclusions

The formalism involved in solving the Schrödinger equation for large-amplitude coupled vibrations on a polynomial potential energy surface has been reviewed. The performance of the PHB and PDGB has been studied and the two bases compared.

The matrix elements of the Hamiltonian are obtained analytically for polynomial potential function and kinetic energy representations in both the harmonic and the DGB basis. In the DGB basis also exponential and Gaussian terms are obtained analytically. In the case of one-dimensional potential functions the calculated levels converge with large precision, both when a harmonic basis and a DGB is used. The precision is larger with the harmonic basis, however. It is fairly straightforward to optimize the parameters of both bases.

The optimal harmonic wavenumber for the harmonic basis depends somewhat on the basis size

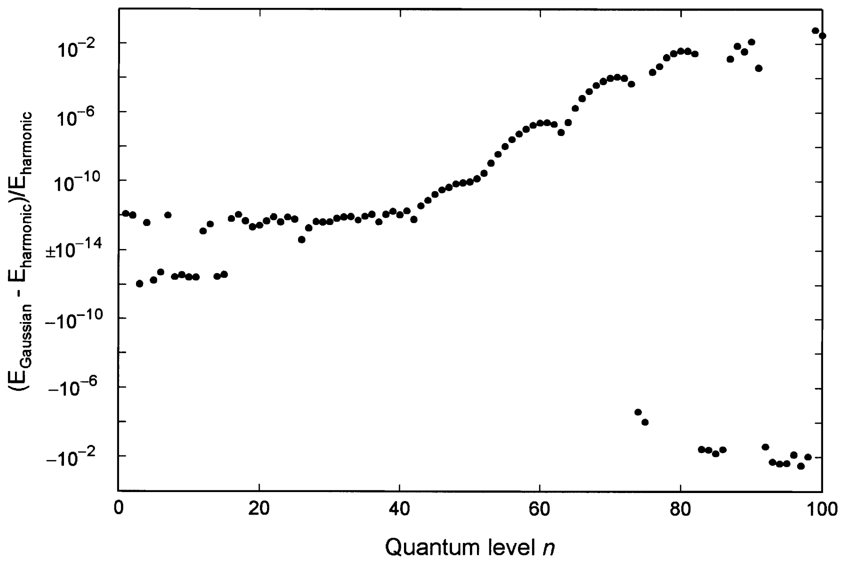

Fig. 11. Comparison of the energy levels of the flapping mode of phthalan calculated with a harmonic basis, $N = 100$, $\bar{\upsilon} = 500$ cm^{-1}, and a Gaussian basis, $N = 100$, $E = 30000$ cm^{-1} ($n = 79$), and $c = 0.50$.

chosen. For the anharmonic potential used in this study it corresponds to the energy level separation in the energy region of the highest converged energy levels. Similar results have been obtained with other potential functions. Between half and one third of the computed levels have converged when the optimum harmonic wavenumber has been chosen. The number of converged levels compared to the basis size depends, of course, on the shape of the potential function, all levels being correctly calculated for a

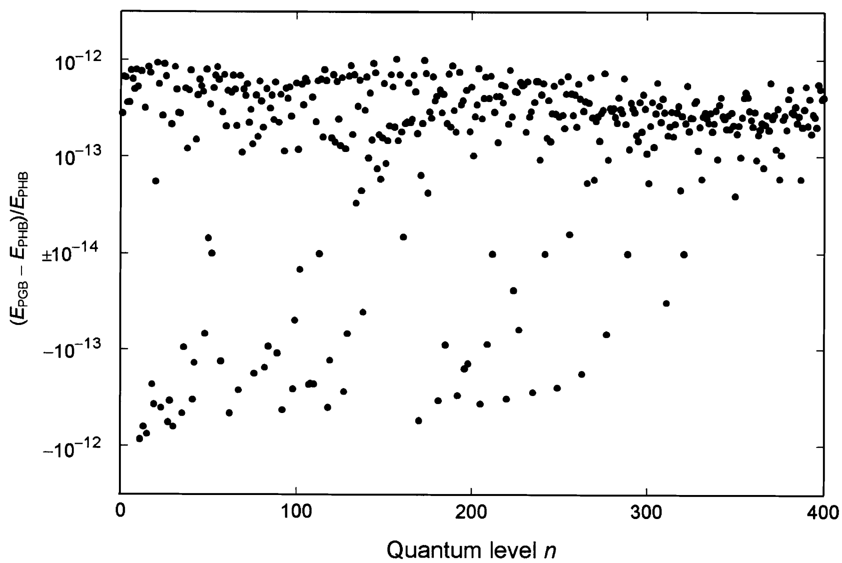

Fig. 12. Comparison of energy levels computed for phthalan with a prediagonalized harmonic basis E_{PHB}, with parameters $N_1 = N_2 = 100$, $\bar{\upsilon}_1 = 325$ cm^{-1}, $\bar{\upsilon}_2 = 325$ cm^{-1}, $N = 25 \times 16$, and a prediagonalized Gaussian basis E_{PGB}, with parameters $c = 0.50$, $N_1 = N_2 = 100$, $E_1 = 20\,000$ cm^{-1} ($n = 82$), $E_2 = 30\,000$ cm^{-1} ($n = 79$), $N = 25 \times 16$.

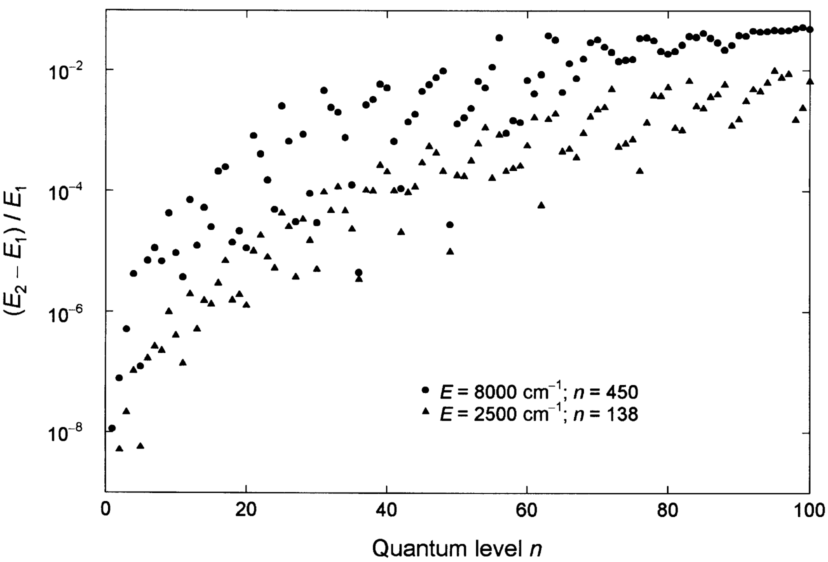

Fig. 13. Comparison of results obtained for phthalan with two two-dimensional Gaussian bases (E_2), of size $N = 716$, $c = 0.5$, $E = 8000$ cm^{-1} corresponding to $n = 450$, and of size $N = 708$, $c = 0.5$, $E = 2500$ cm^{-1} corresponding to $n = 138$, with the result obtained with a prediagonalized Gaussian (E_1) of size 25×16 with parameters $c = 0.5$, $N_1 = N_2 = 100$, $E_1 = 20\,000$ cm^{-1} ($n = 82$), $E_2 = 30\,000$cm^{-1} ($n = 79$).

harmonic potential. The number of converged levels is smaller when a non-optimal harmonic wavenumber is chosen for the basis, but the precision of the convergence does not depend on the harmonic wavenumber of the basis. The calculated energies converge with a precision of the order of 1 part in 10^{14}.

On the order of one half of the calculated energy levels have converged with the DGB if the parameter c and the energy E at which the Gaussians have been positioned are chosen optimally. The optimum value for the parameter c of Eq. (2.31), which determines the overlap of adjacent Gaussians, is $c \approx 0.5$. For this value the Gaussians have decreased to 78% of their amplitude at the centers of the adjacent Gaussians. For smaller values of c, corresponding to larger overlapping of the Gaussians, the errors due to the orthogonalization of the basis increase. For larger values of c the computed energies increase. The precision of the convergence for $c = 0.5$ is on the order of 1 part in 10^{12}. However, for practical applications the results are approximately independent of the choice of this parameter in the range $0.4 < c < 0.8$. Generally the energy E should be at least 20% larger than the highest energy level that is to be computed accurately and the basis size at least twice the number of levels needed.

The energies computed with two-dimensional prediagonalized bases are independent of the parameters of the one-dimensional bases used, provided that the basis functions included in the two-dimensional product basis all correspond to eigenvalues that have converged. The PHB and PDGB bases give identical results, within the precision of the convergence, provided that the product bases contain the same one-dimensional basis functions. The results do, however, depend on the size of two-dimensional basis, showing that convergence has not been reached with the maximum number of 400 basis functions that were used. Nevertheless, the computed energies of the experimentally determined energy levels are accurate enough, even for the much smaller 10×10 basis size often used, not to affect the accuracy of the empirical potential surface models, which are the objective of these computations.

For practical purposes the performance of both the PHB and the PDGB bases are very similar, in terms of computation times, basis set size and precision. The PDGB allows more flexibility in modeling of the potential function, however, as both exponential and Gaussian terms are easily included. Also numerical integration of the matrix elements should be easily implemented for this basis, if the potential function

Table 1
Energy levels for phthalan from one- and two-dimensional calculations

	One-dimensional		Two-dimensional		
	Quantum Level	Energy (cm^{-1})	Quantum Level[a]	Energy (cm^{-1})	Non-interacting[b]
Puckering	$v_P = 0$	36.38	0,0	140.98	150.34
	1	76.62	1,0	174.06	190.58
	2	158.65	2,0	242.67	272.61
	3	253.11	3,0	317.67	367.07
	4	361.31	0,1	360.73	374.29
	5	480.30	1,1	390.47	414.53
	6	608.64	4,0	415.38	475.27
	7	745.24	2,1	464.14	496.56
	8	889.28	5,0	506.88	594.26
	9	1040.12	3,1	514.05	591.02
	10	1197.23	0,2	571.36	590.91
			6,0	606.95	722.60
			1,2	607.33	631.15
			4,1	643.25	699.22
Flapping	v_F	148.85	2,2	682.28	713.18
	1	372.80	7,0	712.34	859.20
	2	589.42	5,1	738.95	818.21
	3	800.11	3,2	755.71	807.64
	4	1006.72	0,3	778.84	801.60
	5	1211.38	1,3	817,46	841.84

[a] (v_P, v_F)
[b] $E(v_P) + E(v_F) - B$ where $B = 34.89\ cm^{-1}$ is the barrier which was taken into account for both one-dimensional calculations.

is modeled piecewise with different analytical functions.

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